

Chapter 06: Structural and Systems Biology:

BIO3107

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What we will learn here!

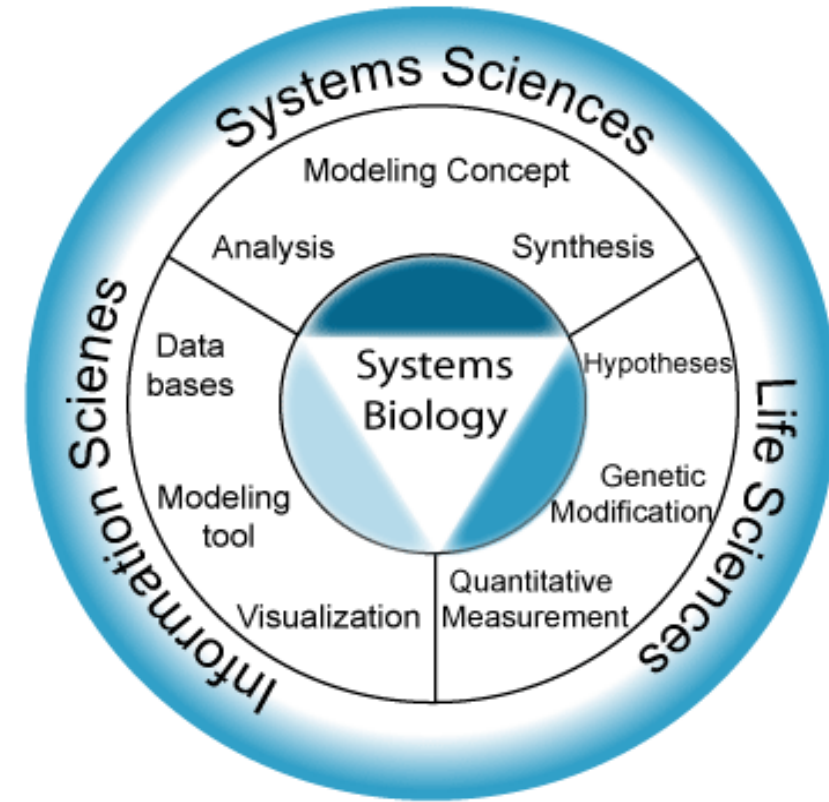
- Big-picture view of biology (biology is complex)
- How engineers study complex systems
- Focus on interactions, not isolated parts
- Many interacting components
- Reductionism is not enough

From Parts to Systems

- Traditional biology: **one gene, one protein**
- Systems biology: networks (“to me, all biology is systems biology”)
- Systems biology is an interdisciplinary approach that *studies complex living systems* as integrated wholes, focusing on the **interactions** between components (like **genes, proteins, cells**) rather than just individual parts using computational models and large-scale data (genomics, proteomics) to understand how these interactions create emergent properties and predict system behavior, moving beyond traditional reductionism to grasp the big picture of biology.

What is a Biological System?

- Set of interacting components
- Genes, proteins, metabolites together contribute to function
- A systems biology approach means:
 1. Investigating the components of cellular networks and their interactions
 2. Applying experimental high-throughput and omics techniques
 3. Integrating computational and theoretical methods with experimental effort



Engineering Analogy

1. Biological Circuits & Control

Cells function like **molecular computers**. Regulatory networks (DNA/proteins) act as logic gates that process information and execute "biological programs." By modeling these as electronic circuits, researchers can predict how a cell will respond.

2. Input-Output & Feedback Loops

Life is governed by information flow:

- **Inputs:** Sensing environmental signals (nutrients, stress, chemicals).
- **Outputs:** Biological actions (growth, movement, hormone release).
- **Feedback:** The system's primary regulator. **Negative feedback** maintains stability (homeostasis), while **positive feedback** creates "switches" for rapid transitions.

3. Robustness & Failure Modes

Evolution "engineers" organisms to be resilient through:

- **Redundancy:** Multiple "back-up" pathways to ensure survival if one gene or protein fails.
- **Modularity:** Isolated functional units prevent a local error from causing a system crash.
- **Failure Modes:** Disease (e.g., cancer) is viewed as a **system-level failure** where regulatory control is lost, requiring system-wide "re-tuning" rather than just a single-point fix.

Key Idea: Networks

Biological Networks: Mapping Life's Architecture

Systems biology uses network theory to visualize and analyze the complexity of life, treating biological systems as structured maps:

- Nodes = Molecules

The "points" on the map. These represent individual biological components, such as genes, proteins, or metabolites. In a disease model, a node might be a specific protein targeted by a drug.

- Edges = Interactions

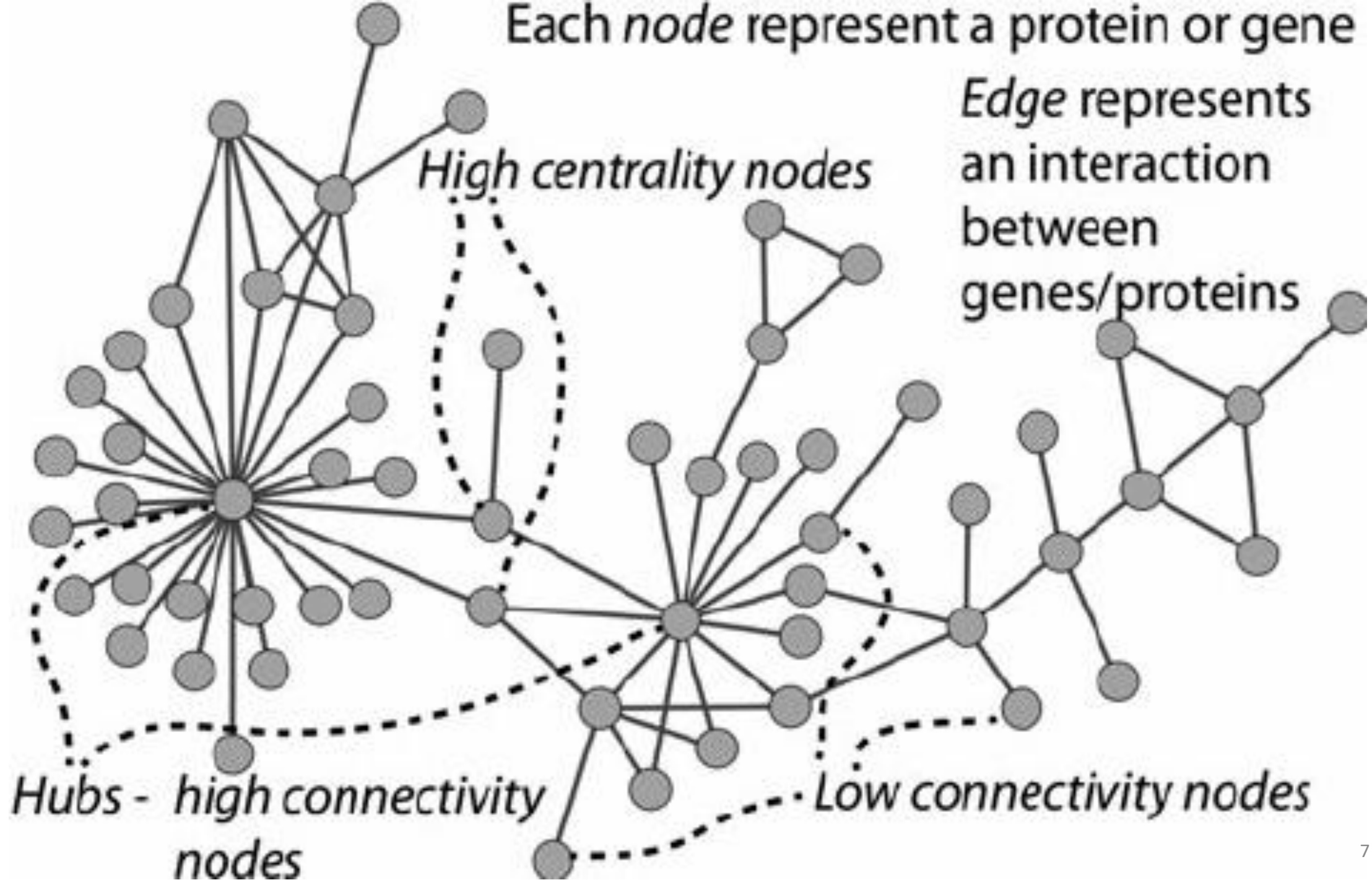
The "lines" connecting the nodes. These represent the relationships between molecules, such as a protein binding to DNA, a chemical reaction, or physical signaling. The "weight" (thickness) of an edge often indicates the strength of that interaction.

By combining nodes and edges, researchers create a "Interactome" map. This visual blueprint allows scientists to identify "hubs" (critical molecules) and "bottlenecks," transforming abstract data into a navigable diagram for identifying drug targets and understanding system behavior.

Each *node* represent a protein or gene

Edge represents
an interaction
between
genes/proteins

High centrality nodes



*Hubs - high connectivity
nodes*

Low connectivity nodes

Types of Biological Networks

- Metabolic networks
- Gene regulatory networks
- Signaling networks

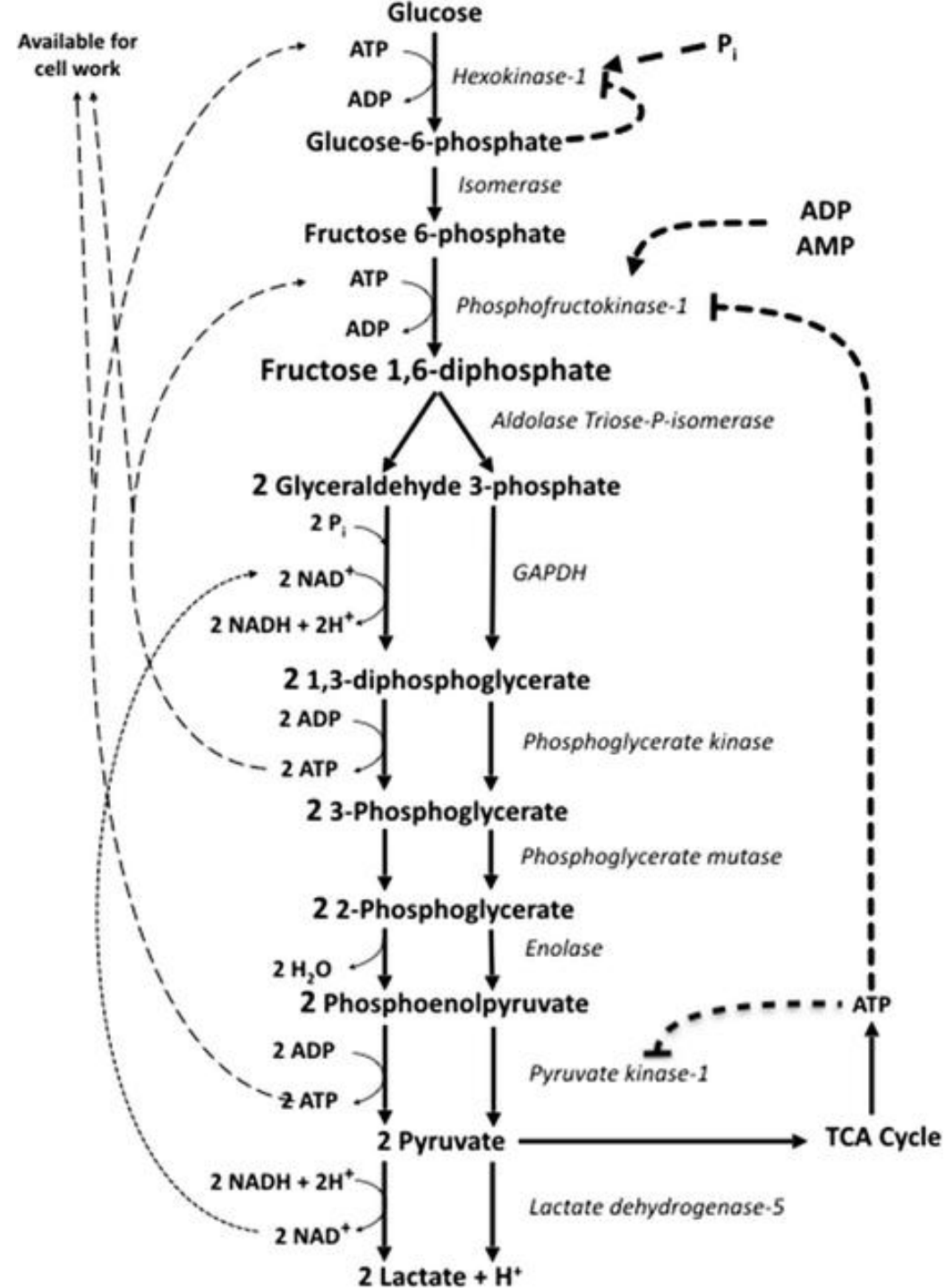
Metabolic Networks

- Metabolism: defined as the sum of chemical reactions
- Contributes to synthesizing energy and building blocks
- Example: glycolysis

(Nodes = Metabolites: These represent the small molecules (substrates and products) involved in chemical reactions, such as glucose, ATP, or pyruvate.)

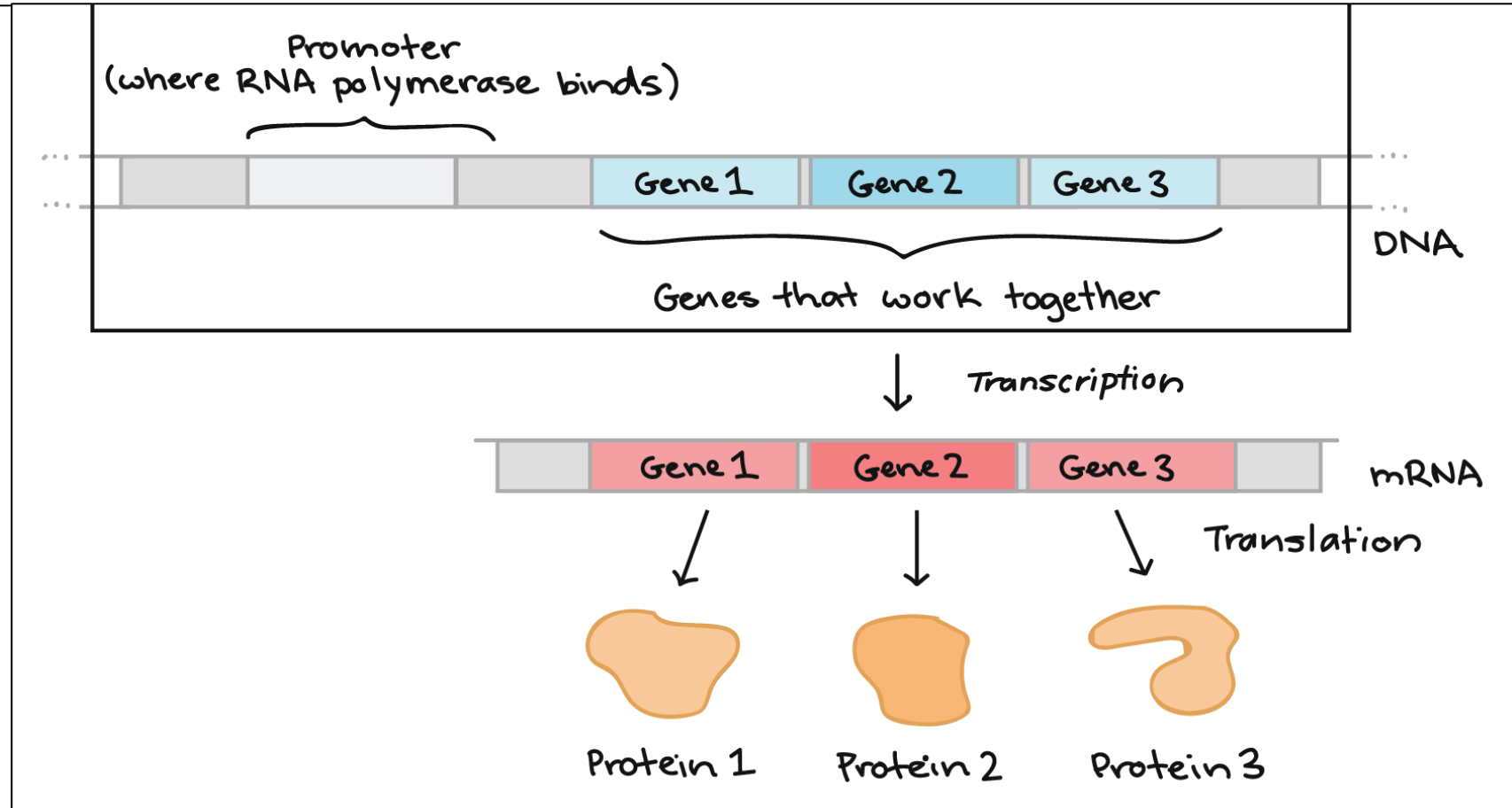
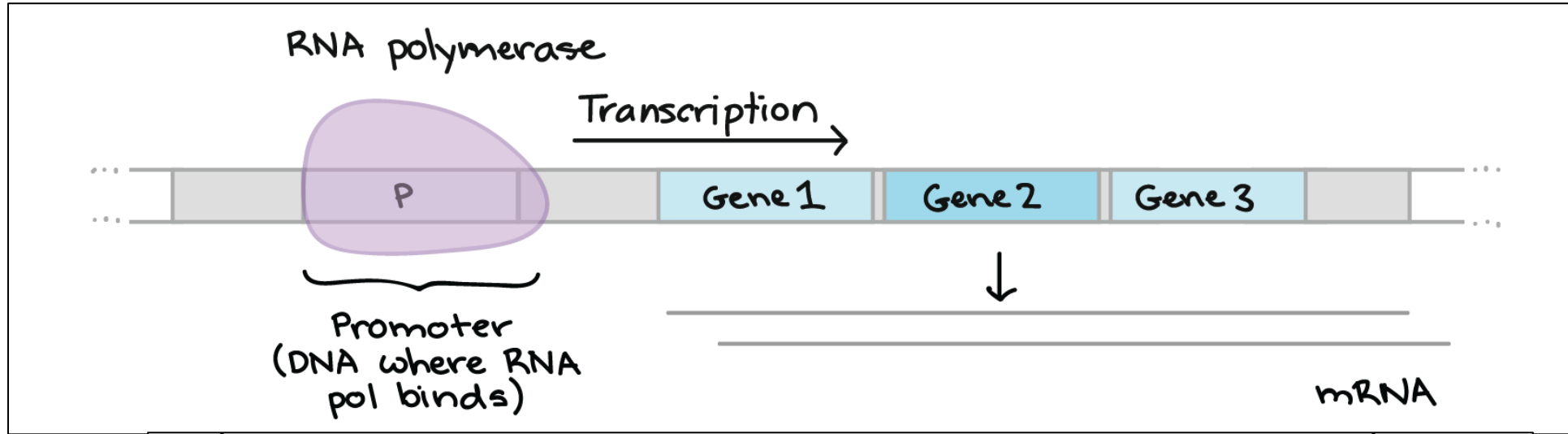
(Edges = Reactions: The lines connecting nodes represent the biochemical transformations themselves, which are almost always facilitated by enzymes.)

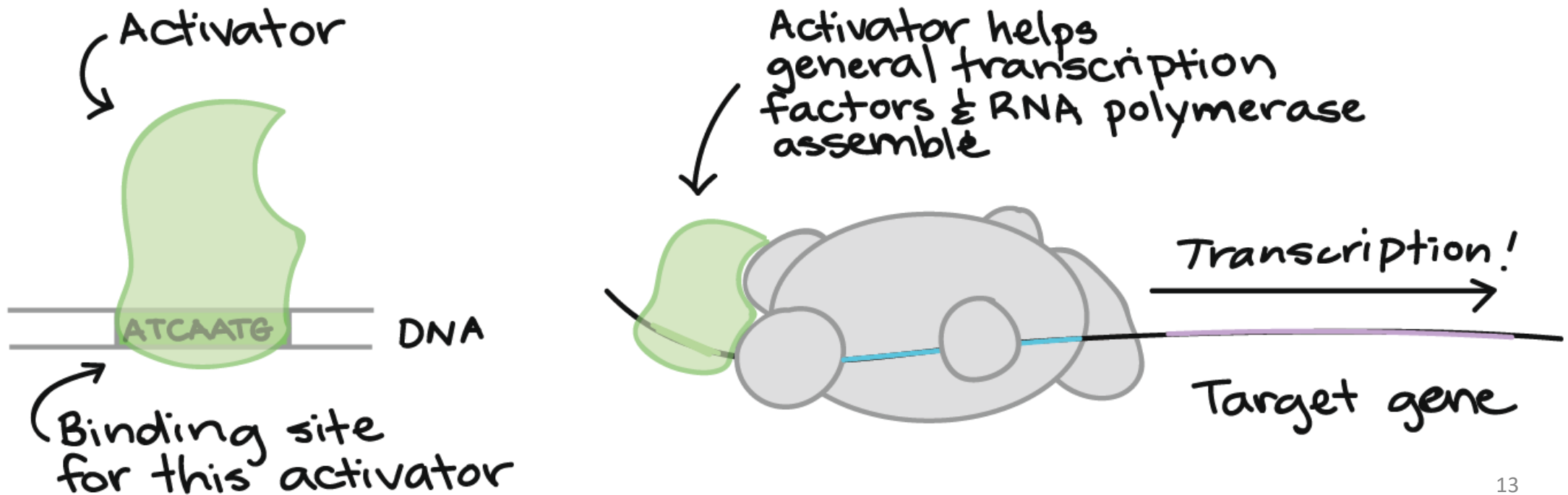
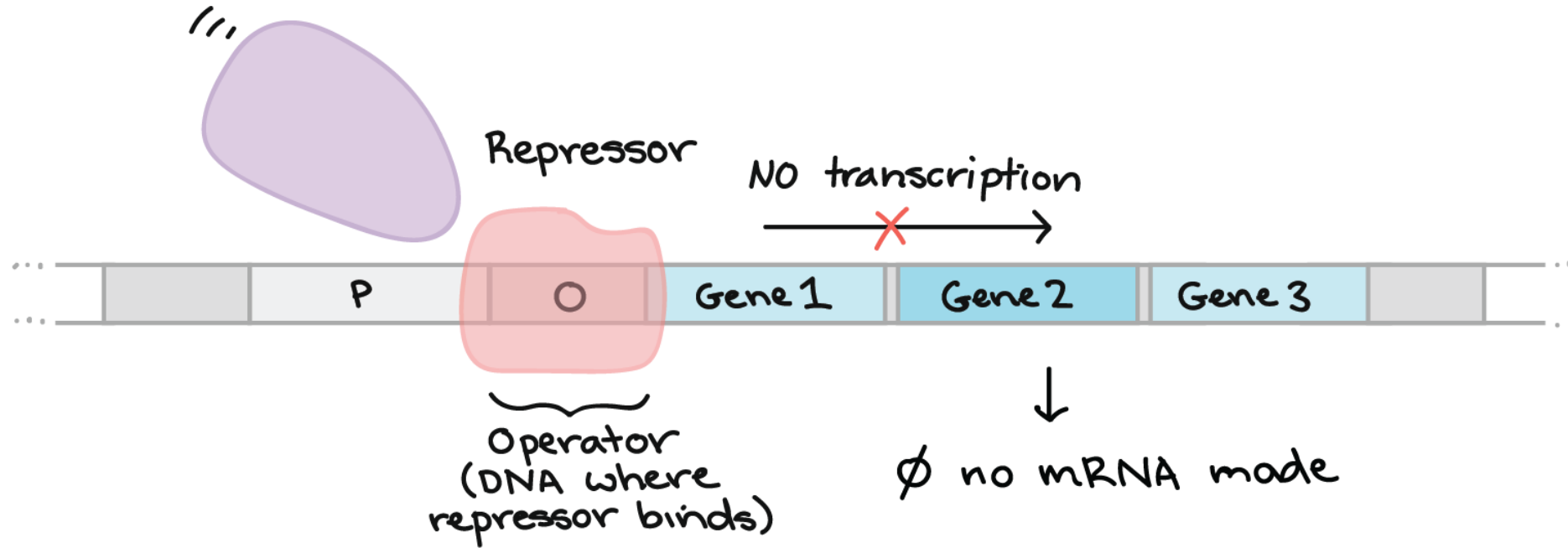
(Directionality: Edges are often "directed" (arrows) to indicate the flow of matter from reactants to products)



Gene Regulatory Networks

- Genes control and regulate other genes (on/off logic)
- **Nodes = Genes & Proteins:** Specifically, **Transcription Factors (TFs)**, proteins that act as master controllers, and the target genes they regulate.
- **Edges = Regulatory Links:** The physical binding of a TF to DNA to either activate ("turn on") or repress ("turn off") a gene.





Signaling Networks

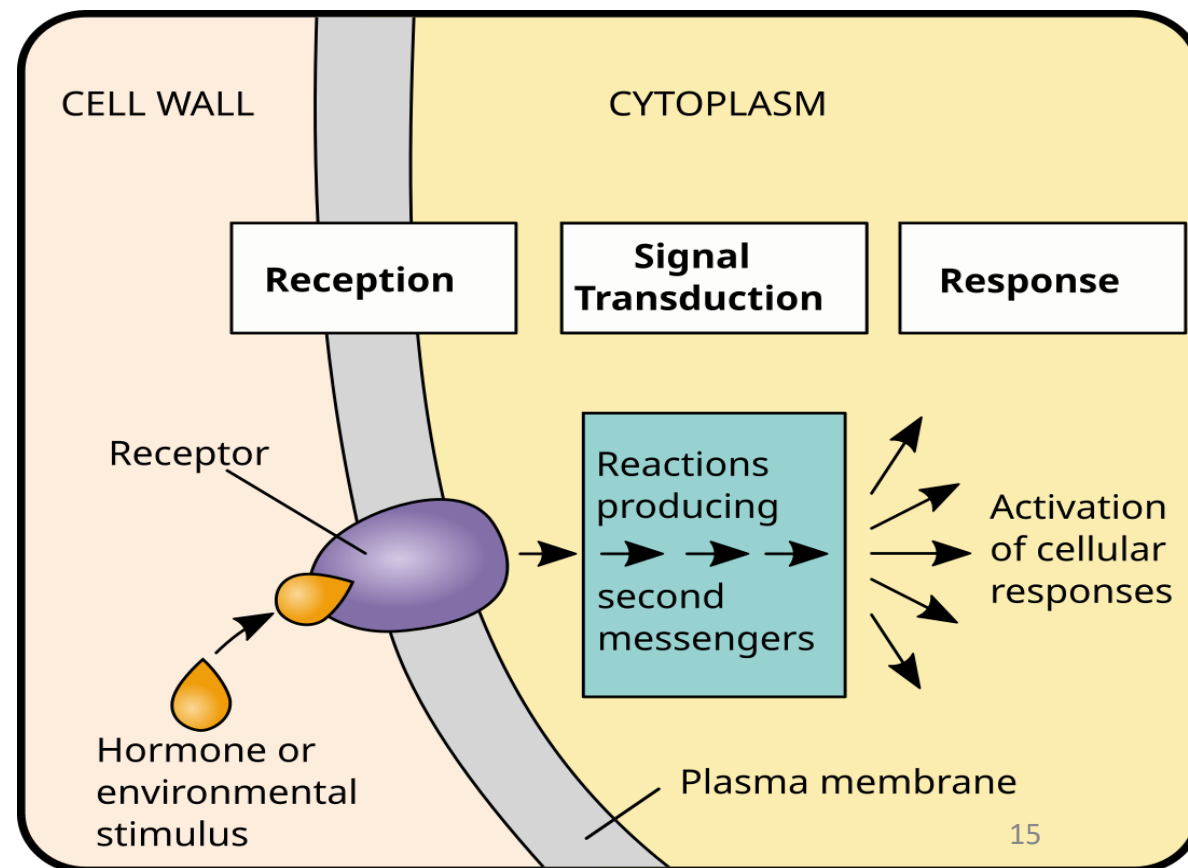
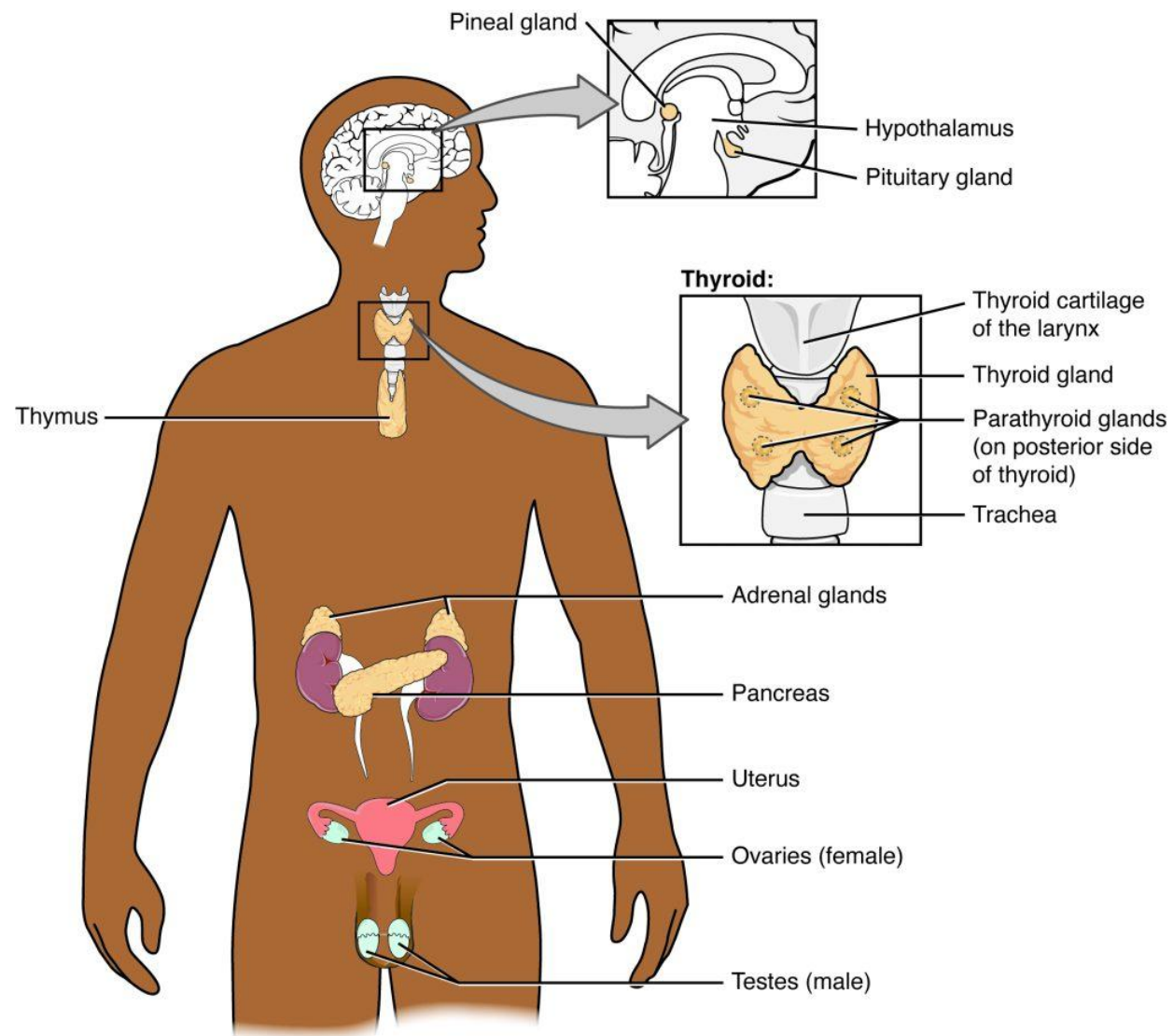
Signaling networks are mapped as the cell's "communication circuitry" that translates environment into action.

- **Responsive Sensors:** Cells detect environmental stimuli via surface receptors (nodes), initiating a relay cascade that carries "messages" from the membrane to the nucleus.
- **The Inputs (Hormones & Growth Factors):**

Hormones: Long-range messengers (e.g., Insulin) for system-wide coordination.

Growth Factors: Local signals that "instruct" cells to divide or heal; often the targets of modern precision oncology.

- **Information Processing:** The network acts as a biological CPU. It integrates multiple conflicting inputs, amplifies weak signals into strong actions, and uses crosstalk between pathways to "calculate" the optimal survival response.



Core Features of Complex Systems

- Nonlinearity: A small change in input can lead to a disproportionately large shift in the output.

(Engineering Analogy: Tipping points and phase transitions. For example, in cancer, slow changes accumulate linearly until the system abruptly "switches" into a malignant, aggressive growth state. The behavior cannot be predicted by simple linear extrapolation.)

- Feedback loops
- Robustness

Feedback Loops

1. Common in biology
2. System outputs loop back to influence their own inputs, creating self-regulating cycles. These are the engines of biological control.
3. **Engineering Analogy:** Control systems and regulation.
 - **Negative Feedback:** stabilizes systems by reducing or reversing a stimulus. Crucial for maintaining stability, like in the case of **homeostasis** (e.g., keeping body temperature stable).
 - **Positive Feedback:** amplifies a stimulus, pushing the system further from its set point for rapid, often temporary events. Drives amplification or rapid, "all-or-nothing" **switch-like behavior** (e.g., initiating cell division or a nerve impulse)

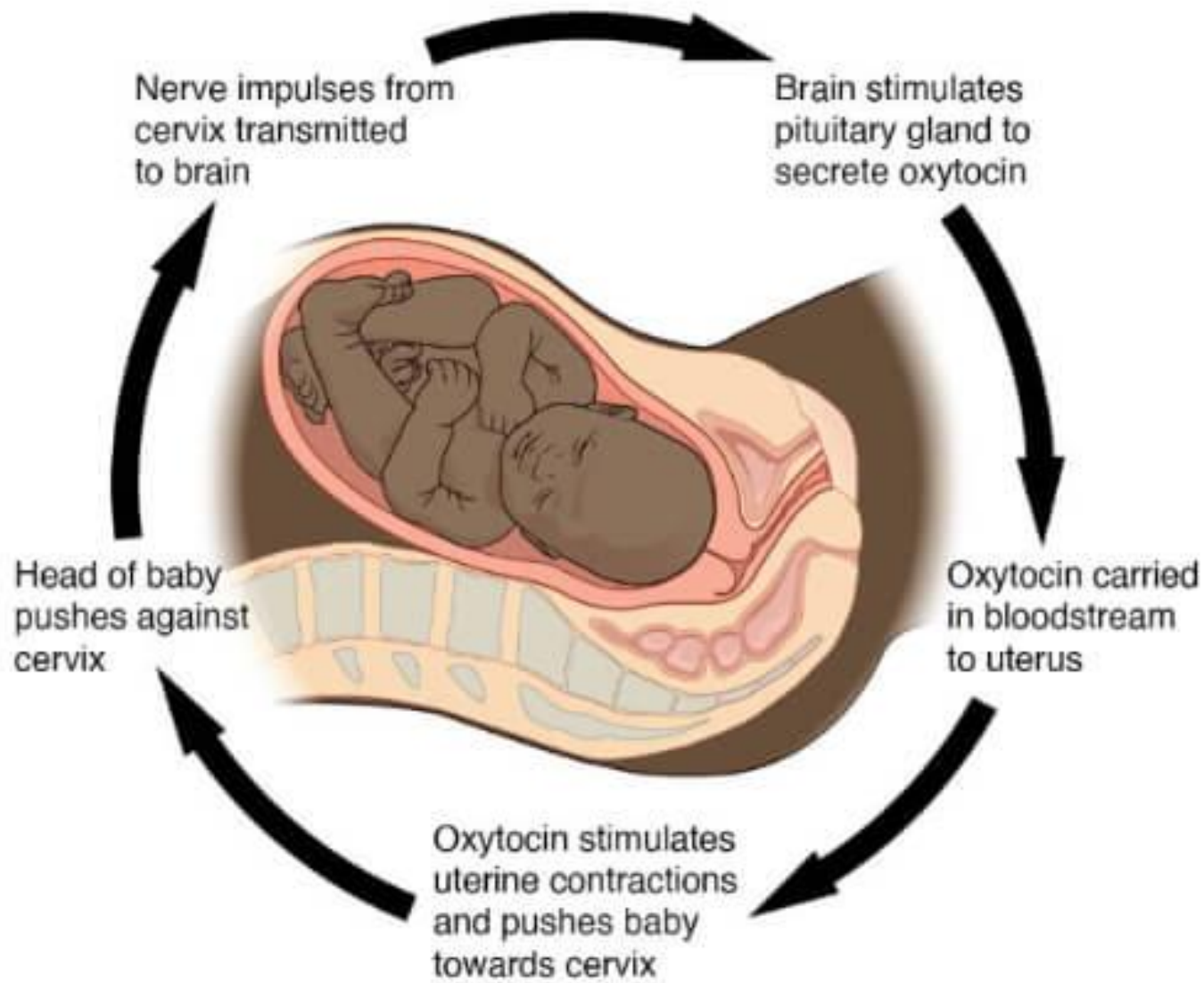


Fig: **Positive Feedback** (Childbirth)

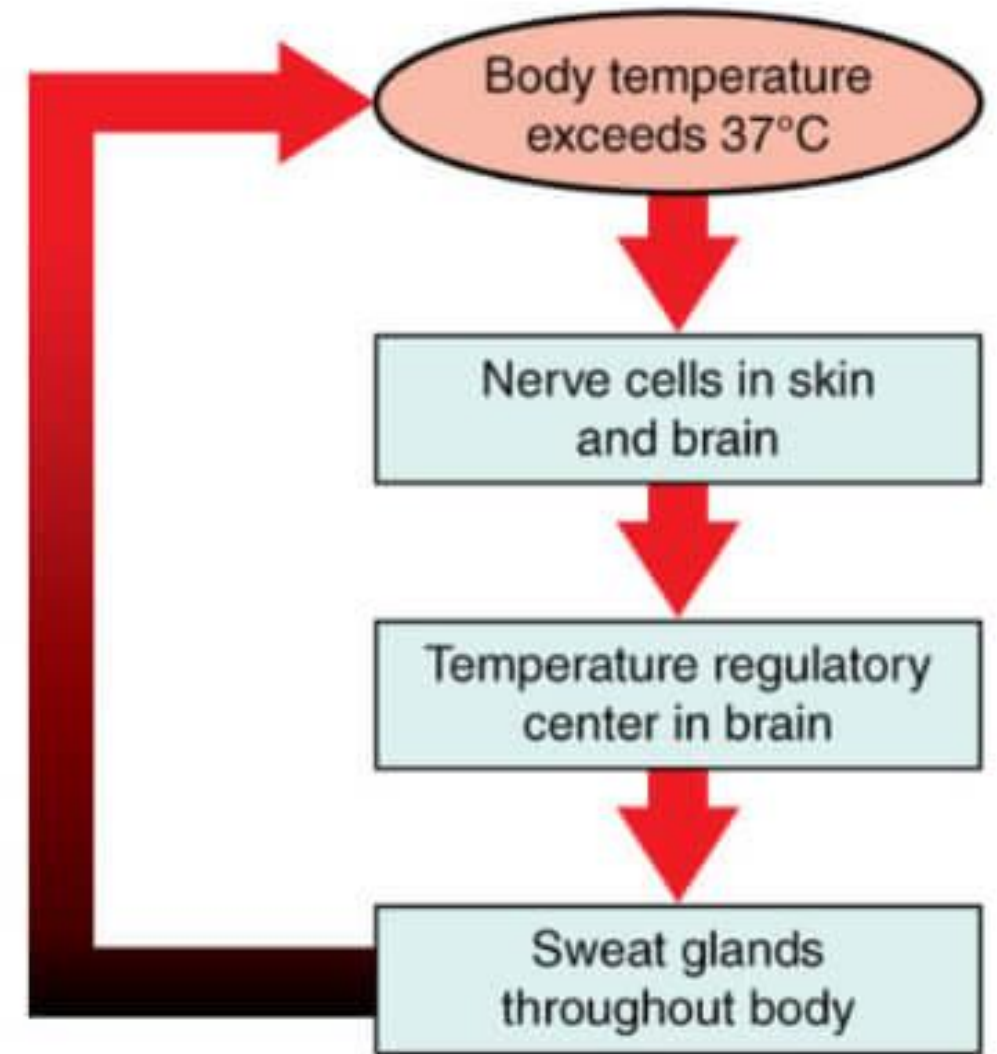


Fig: **Negative Feedback**
(Temperature Regulation)

Robustness

- Robustness is analyzed as the biological equivalent of "fail-safe engineering," ensuring that life continues even when conditions are far from perfect.
- Noise Filtering: Systems use regulatory loops to ignore molecular "static," ensuring stable behavior despite random fluctuations.
- Environmental Tolerance: Performance remains steady even when external parameters (temp, pH, nutrients) shift, similar to mechanical cruise control.
- Fault-Tolerant Design: Like a power grid, biology uses redundancy (backups) and modularity (containment) to ensure that a single component failure doesn't lead to a total system collapse.

Modeling in Systems Biology

- Models = simplified representations
- Help explain and predict behavior
- Not exact copies

Why Mathematical Models?

- Scale and randomness determine how we model biological systems. Hence, math drives experimental and model design.
- Quantitative Predictions & Design: We move beyond "protein A activates B" to "protein A must reach a concentration of 10nM to trigger B." These numbers allow us to design experiments with specific thresholds in mind, reducing lab waste.
- Hypothesis Testing (The Loop): Modeling is a circular process. We build a model, run a simulation, compare it to 2026 real-world data, and if they don't match, we know a biological "part" is missing from our map.
- The Modeling Spectrum:
 1. Mechanistic: Deep-dive "wiring diagrams." Used to find drug targets because every parameter represents a physical molecule.
 2. Phenomenological: High-speed statistical forecasting. Excellent for predicting cell growth rates under different nutrient conditions without needing to map every single enzyme.

Biological randomness is a feature, not a bug

- Deterministic Models (The Machine): When looking at millions of cells (like an organ), we use Ordinary Differential Equations (ODEs). At this scale, individual variations average out, and the system behaves like a predictable machine.
- Stochastic Models (The Gamble): At the single-cell level, molecule numbers are so low that random collisions dictate behavior. We use probabilistic models to account for this Intrinsic Noise.

Why Noise Matters (noise refers to inherent random fluctuations in biological processes, from gene expression to cellular behavior):

1. Cell Fate Decisions: Random "noise" in gene expression is often what triggers a stem cell to decide whether to become a heart cell or a lung cell.
2. Bet-Hedging: Bacteria use noise to ensure that in a population, a few cells are always "weirdly" resistant to antibiotics, even if they haven't been exposed to them yet.
3. Failure Modes: researchers look at how "too much noise" in signaling can lead to age-related diseases or cellular malfunctions.

Experimental Data for Systems Biology

- High-throughput measurements
- Large datasets (omics technologies)
- Time-series data

Omics Technologies

- Genomics
- Transcriptomics
- Proteomics
- Metabolomics

Systems Biology Workflow

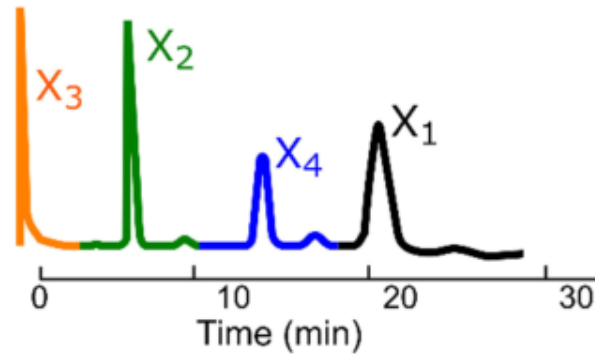
- Measure → Model → Predict → Test
- Iterative cycle
- Engineering mindset

Metabolomics

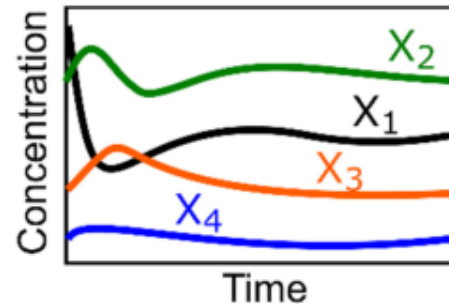
Biological Samples



Data Acquisition



Data Processing

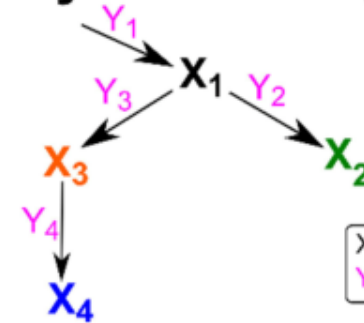


Mathematical modelling

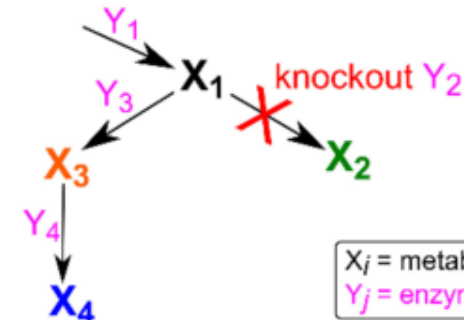
Model Construction

$$\frac{dX_i}{dt} = \sum_{j=1}^p \alpha_{ij} \prod_{k=1}^n X_k^{g_{ijk}} - \sum_{j=1}^q \beta_{ij} \prod_{k=1}^m X_k^{h_{ijk}}$$

System Analysis



System Design



Future Directions

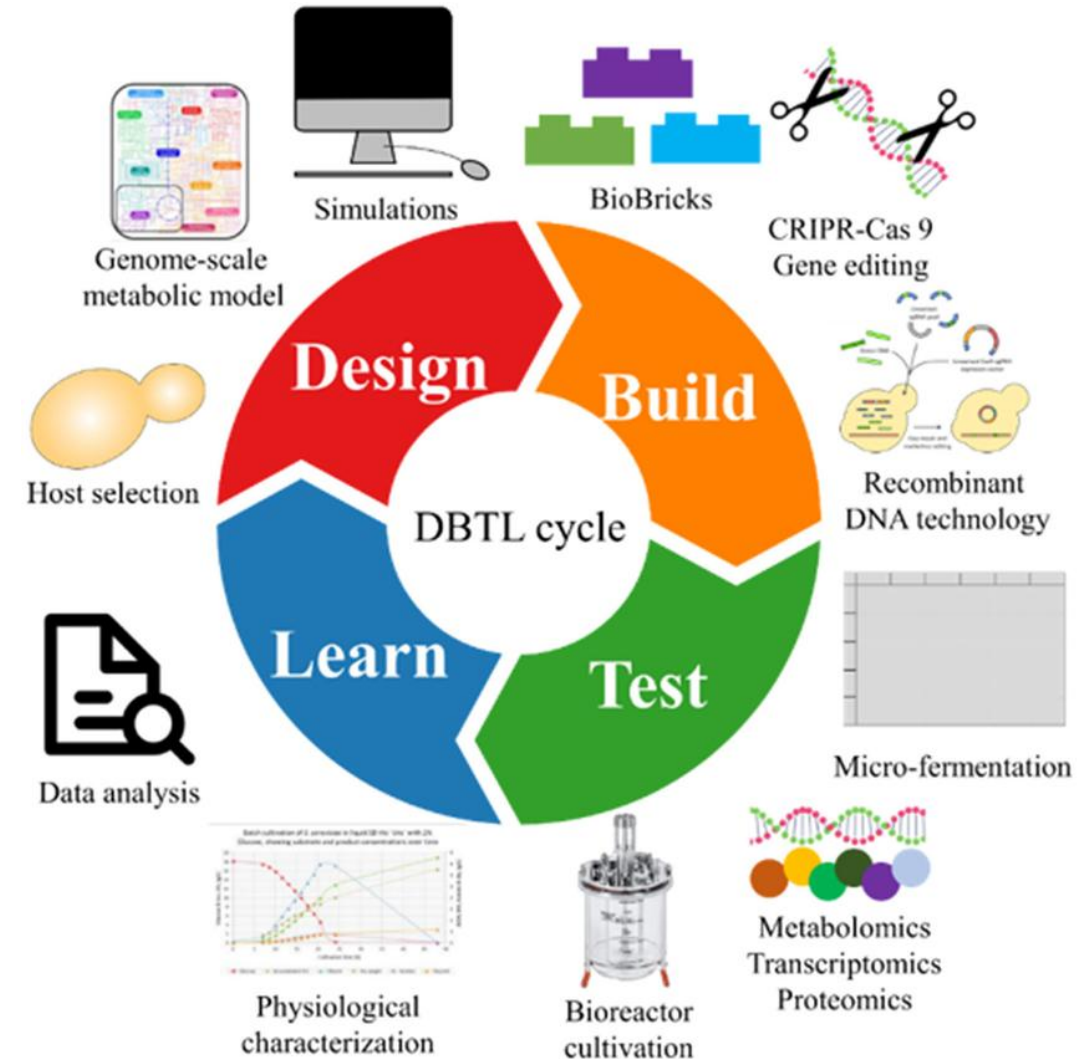
- Synthetic biology
- Personalized medicine
- Cellular engineering

Synthetic Biology

- Synthetic Biology is the design and construction of new biological parts, devices, and systems, and the re-design of existing, natural biological systems for useful purposes
- Synthetic biology treats the cell like a computer and DNA like software code.
- **Hardware** : This is the host cell (like *E. coli* bacteria or Yeast). It provides the energy, the raw materials, and the machinery to run the code.
- **Software (The Genetic Circuit)**: This is the synthetic DNA you design and insert. It gives the instructions.
- **Design-Build-Test-Learn (DBTL) cycle** is the core engineering loop in synthetic biology for developing new biological systems.

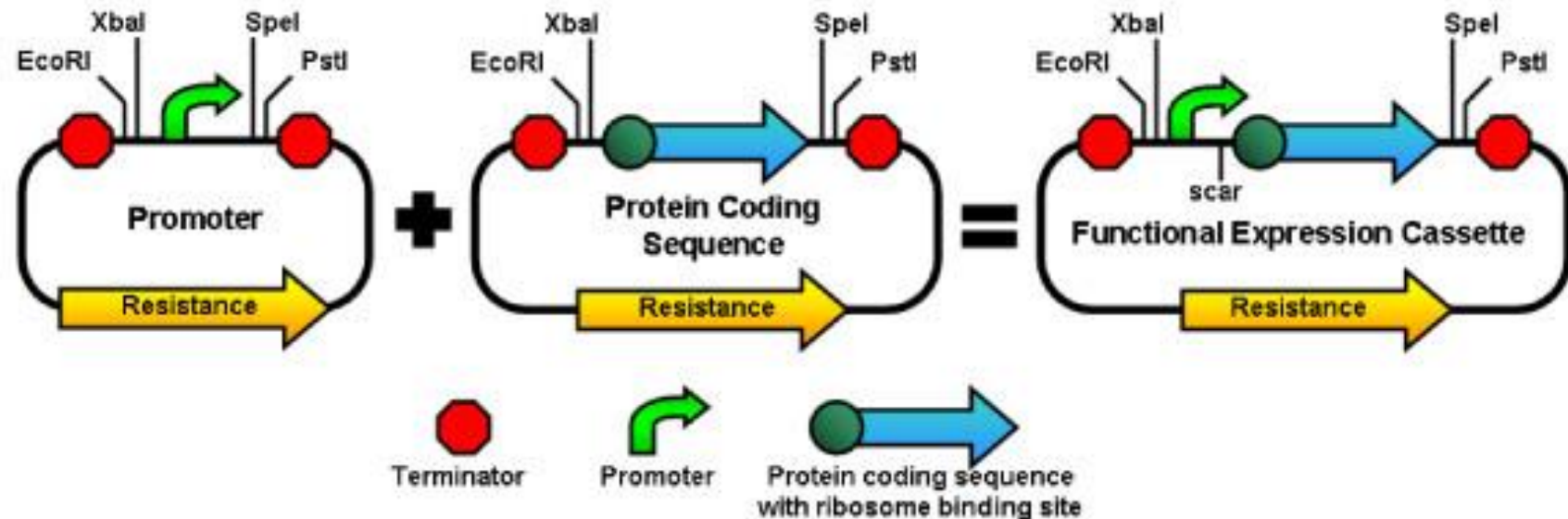
Synthetic Biology

- Design:** Scientists use computational tools and bioinformatics to plan genetic circuits and metabolic pathways to achieve a desired biological function.
- Build:** Digital designs are converted into physical DNA using synthesis, standardized BioBricks, and gene editing tools like CRISPR-Cas9 to create engineered organisms.
- Test:** These organisms are cultivated and evaluated using high-throughput screening and multi-omics data collection to measure their performance.
- Learn:** Data is analyzed, often using AI and machine learning, to refine models and gain insights that inform the next design iteration.



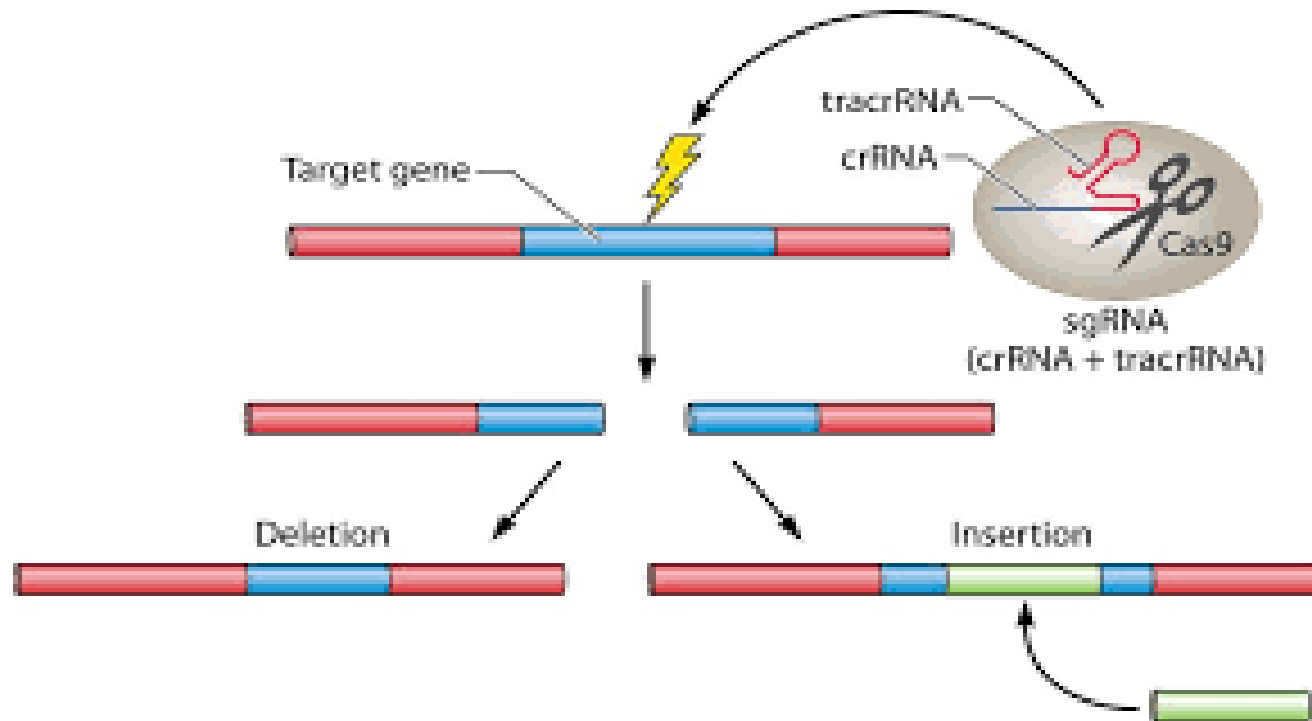
Core Tools & Techniques in Synthetic Biology

- DNA synthesis:** Scientists chemically create DNA sequences in the lab instead of copying them from nature. This allows precise design of genes, promoters, and regulatory elements for specific functions.
- BioBricks:** BioBricks are standardized DNA parts (such as promoters, coding sequences, and terminators) that can be mixed and matched like LEGO blocks to build genetic circuits.



Core Tools & Techniques in Synthetic Biology

- CRISPR–Cas9:** It is a powerful gene-editing tool adapted from a bacterial defense system.
- How it works: A single guide RNA (sgRNA) directs the Cas9 enzyme to a specific DNA sequence, where Cas9 cuts the DNA. The cell repairs the cut, allowing genes to be deleted, corrected, or new DNA to be inserted.
- It enables fast, precise, and low-cost genome editing in research, medicine, and synthetic biology.



Applications of Synthetic Biology

- Biomedicine:** Modified cells are used to fight diseases, like cancer treatment and making vaccines quickly.
- Biofuels:** Special microbes turn waste or algae into cleaner fuels.
- Biosensors:** Living cells are made to detect harmful chemicals or viruses and show signals like color or light.
- Food ingredients:** Lab-grown meat and microbes make milk or meat proteins without animals.
- Bioremediation:** Engineered microbes help clean pollution such as oil spills, plastics, and toxic metals.

